Day 51 + Day 52

Model Monitoring, Maintenance, Automated Retraining and ML Pipelines

Learning outcomes

Explain model monitoring, distinguish data drift from concept drift, evaluate regression errors, build a Scikit-learn pipeline, compare model versions, and save an approved model with monitoring records.

Study order: Revise Day 51 first, then study Day 52. Run the companion Python practical after installing its dependencies.

DAY 51. MODEL MONITORING AND MAINTENANCE

1. What is model monitoring?

Model monitoring is the ongoing process of checking a deployed machine learning system. It helps detect changing data, poor predictions, service errors, slow response times, and other operational problems. A model that performed well during testing can become less useful as real-world conditions change.

Example: A house-price model trained on last year's property market may become less accurate after changes in interest rates, transport, demand, or construction costs.

2. What should you monitor?

Area	What to check
Input quality	Missing values, invalid ranges, unexpected categories
Data distribution	Feature means, spread, and category proportions
Prediction behavior	Output ranges and unusual prediction patterns
Model quality	MAE, RMSE, R^2, precision, recall, or F1 when labels arrive
Operations	Latency, errors, throughput, and availability

3. Data drift, concept drift, and prediction drift

Data drift: The distribution of incoming input features changes. Example: incoming customer ages shift from mostly 20–40 to mostly 45–65.

Concept drift: The relationship between input features and the target changes. Example: a new transport route changes how strongly location affects property prices.

Prediction drift: The distribution of model outputs changes. This is a signal to investigate, not proof that the model is wrong.

4. Model evaluation metrics

MAE is the average absolute difference between actual and predicted values. Lower MAE means smaller average absolute errors. RMSE penalizes large errors more strongly. R^2 describes model fit relative to a baseline. For classification, common metrics include accuracy, precision, recall, F1-score, and a confusion matrix.

```
MAE = mean(abs(actual - predicted))
```

5. Responding to monitoring alerts

Inspect data quality, check whether the data source changed, verify labels and metric calculations, compare against a stable baseline, and decide whether retraining is justified. A drift alert does not automatically mean model performance has decreased. Thresholds must come from validation history and business requirements.

DAY 52. AUTOMATED RETRAINING AND ML PIPELINES

6. What is model retraining?

Retraining means fitting a model again using updated, relevant, quality-checked data. It may be scheduled, triggered by an event, or started manually. Evaluate the new candidate before replacing an approved model.

7. When should you retrain?

Consider retraining when reliable new data is available, model quality falls below an agreed level, concept drift affects the target relationship, or the business problem changes. Do not retrain blindly after every drift alert. Investigate first.

8. What is an ML pipeline?

A pipeline connects preprocessing and model training into a repeatable sequence. Scikit-learn Pipeline keeps transformations and the estimator together. When fitted on training data, preprocessing learns from training data only, helping reduce leakage and ensuring consistent transformations at prediction time.

```
Raw data -> preprocessing -> model training\n          -> evaluation -> acceptance decision\n          -> save approved
```

9. Why save the complete pipeline?

Saving the complete pipeline preserves preprocessing and the estimator. At prediction time, the loaded artifact applies the same transformations before generating predictions. This reduces training-serving inconsistency.

10. Candidate model approval

Compare the candidate and current model using a suitable validation strategy. Define acceptance criteria before looking at results. A candidate may need to improve MAE and meet a business-defined error limit. Consider latency, robustness, and fairness where relevant. If it fails, retain the previous approved model and preserve version history for rollback.

11. MLOps vocabulary

Term	Meaning
Model registry	Tracks model versions and lifecycle stages
Experiment tracking	Records parameters, metrics, datasets, and artifacts
CI/CD	Automates testing and delivery or deployment
Orchestration	Coordinates scheduled and dependent workflow steps
Rollback	Returns to a previously approved model or release

12. Tools to explore

MLflow supports experiment tracking and model lifecycle management. Apache Airflow schedules and orchestrates workflows. Kubeflow supports ML workflows on Kubernetes. GitHub Actions automates repository checks and deployment tasks. Understand the workflow before adopting complex platforms.

COMBINED PRACTICAL PROJECT

13. ML Model Monitoring and Retraining System

The companion Python project loads Scikit-learn's diabetes regression dataset, splits the data, trains a baseline pipeline, checks for a simple feature mean shift, trains a candidate model, compares MAE, saves the approved artifact, and writes monitoring and drift CSV files.

Install and run

```
pip install numpy pandas scikit-learn joblib\npython Day-51-52-ML-Monitoring-Retraining-Project.py
```

14. Generated outputs

- day51_52_outputs/approved_model.joblib, the selected complete pipeline.
- day51_52_outputs/monitoring_log.csv, timestamp, version, MAE, status, and drift count.
- day51_52_outputs/drift_report.csv, per-feature mean-shift comparison.

15. Understand the drift check

The project compares each feature's incoming mean with its reference mean, scaled by the reference standard deviation. This is a simple teaching demonstration, not a complete drift detector. It cannot identify every distribution change and does not prove predictive performance has fallen. Production systems may use statistical tests, distribution distances, domain limits, and labeled performance metrics.

16. Understand the model comparison

The script compares a baseline Random Forest with 50 trees against a candidate with 150 trees. It uses the same held-out test split for teaching simplicity. A real workflow should use training, validation, and final test sets separately. Avoid repeatedly tuning against the final test set.

17. Project tasks and solution criteria

Task	Correct solution
Train baseline	Fit preprocessing and estimator on training data.
Calculate MAE	Compare actual and predicted values on held-out data.
Check drift	Compare reference and incoming feature distributions.
Train candidate	Fit a new pipeline using validated training data.
Compare models	Apply pre-defined acceptance rules on validation data.
Save approved model	Persist the complete pipeline and retain prior version if rejected.
Log monitoring	Record timestamp, version, metric, status, and drift count.

18. Combined quiz with answers

Q1. What is model monitoring?
Checking a deployed model and its data and operational behavior over time.

Q2. What is data drift?
A change in the distribution of incoming input data.

Q3. What is concept drift?
A change in the relationship between model inputs and target.

Q4. What is retraining?
Fitting a model again using updated and validated data.

Q5. What is an ML pipeline?
A repeatable sequence of preprocessing and model steps.

Q6. Why compare a candidate model?
To check it meets acceptance criteria before replacing the approved model.

Q7. What does rollback mean?
Returning to a previously approved model or release.

Q8. Does drift always mean performance is worse?
No. Drift is a signal to investigate. Labeled outcomes are needed to measure actual predictive performance.

19. Production checklist

- Validate new data before training.
- Keep training, validation, and final test data separate.
- Define acceptance criteria before model comparison.
- Track datasets, code, parameters, metrics, and model versions.
- Protect model artifacts and sensitive data.
- Monitor latency, errors, and prediction quality.
- Keep the previous approved model for rollback.
- Add a human approval gate where impact is significant.

Dataset note: The Scikit-learn diabetes dataset is used only to demonstrate regression workflows. It is not suitable for diagnosis, treatment decisions, or clinical use.

GitHub learning repository: https://github.com/DeveloperDesigner-star/Datasciencepdfs